

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'The best type of laser with which to do spectroscopy over a range of visible wavelengths is' is 'an excimer laser'.
2. True or False: The correct answer to 'The rest mass of a particle with total energy 5.0 GeV and momentum 4.9 GeV/c is approximately' is ' $1.0 \text{ GeV}/c^2$ '.
3. True or False: The correct answer to 'A rod measures 1.00 m in its rest system. How fast must an observer move parallel to the rod to measure its length to be 0.80 m?' is ' $0.60c$ '.
4. True or False: The correct answer to 'A single-electron atom has the electron in the $l = 2$ state. The number of allowed values of the quantum number m_l is' is '5'.
5. True or False: The correct answer to 'Which of the following is true about any system that undergoes a reversible thermodynamic process?' is 'There are no changes in the internal energy of the system.'
6. True or False: The correct answer to 'The sign of the charge carriers in a doped semiconductor can be deduced by measuring which of the following properties?' is 'Hall coefficient'.
7. True or False: The correct answer to 'The energy required to remove both electrons from the helium atom in its ground state is 79.0 eV. How much energy is required to ionize...' is '51.8 eV'.
8. True or False: The correct answer to 'Electromagnetic radiation emitted from a nucleus is most likely to be in the form of' is 'ultraviolet radiation'.
9. True or False: The correct answer to 'A refracting telescope consists of two converging lenses separated by 100 cm. The eye-piece lens has a focal length of 20 cm. The angular...' is '4'.
10. True or False: The correct answer to 'Which of the following is true about any system that undergoes a reversible thermodynamic process?' is 'There are no changes in the internal energy of the system.'

End of Paper